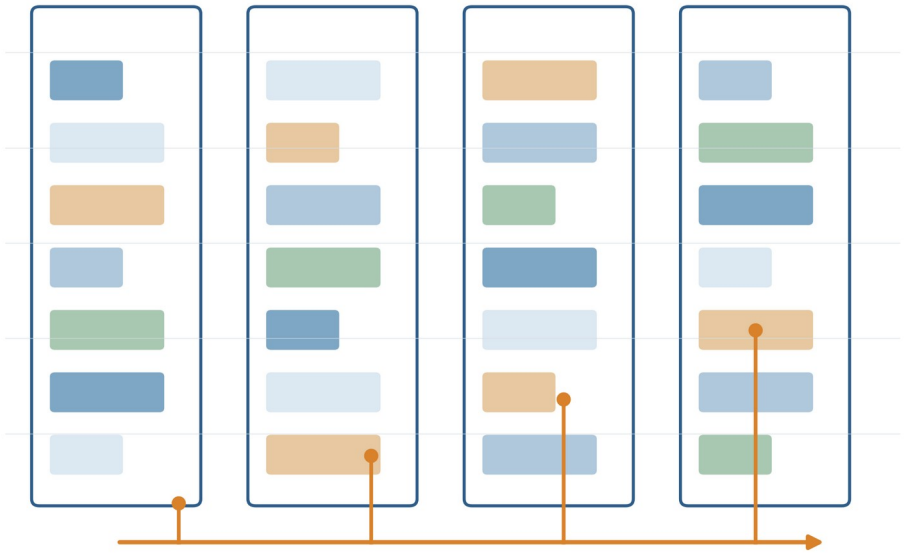
Revisiting Lifetime-Aware VM Allocation

How prediction quality changes server packing

Can better lifetime estimates help when predictions are wrong?



CE7490 Course Project · 26S1 Group 5

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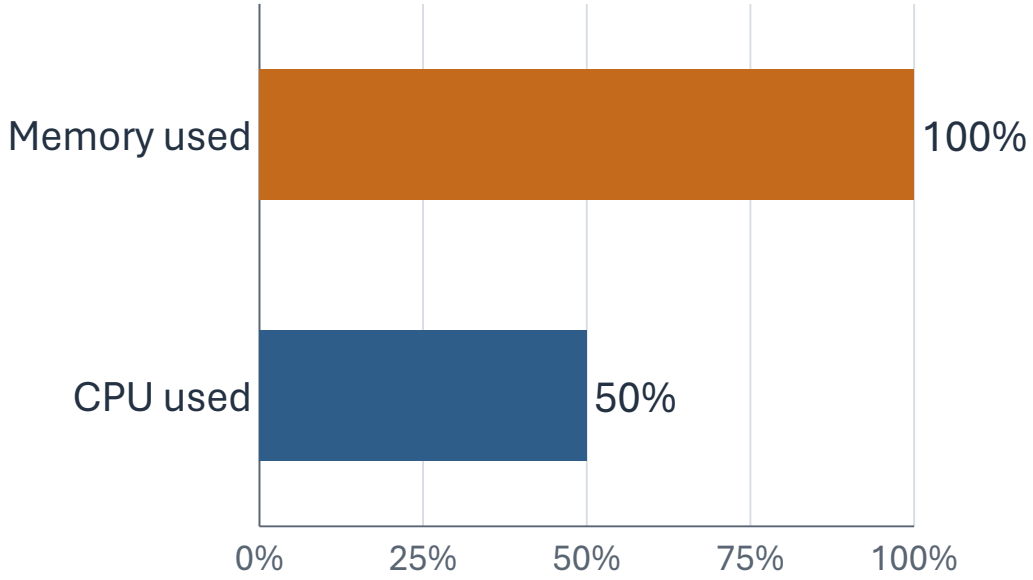
Virtual machines and physical hosts

VM = a small virtual computer

Host = the physical server
that runs several VMs

A new VM needs both CPU and memory.
Free CPU alone is not enough.

A full-memory example

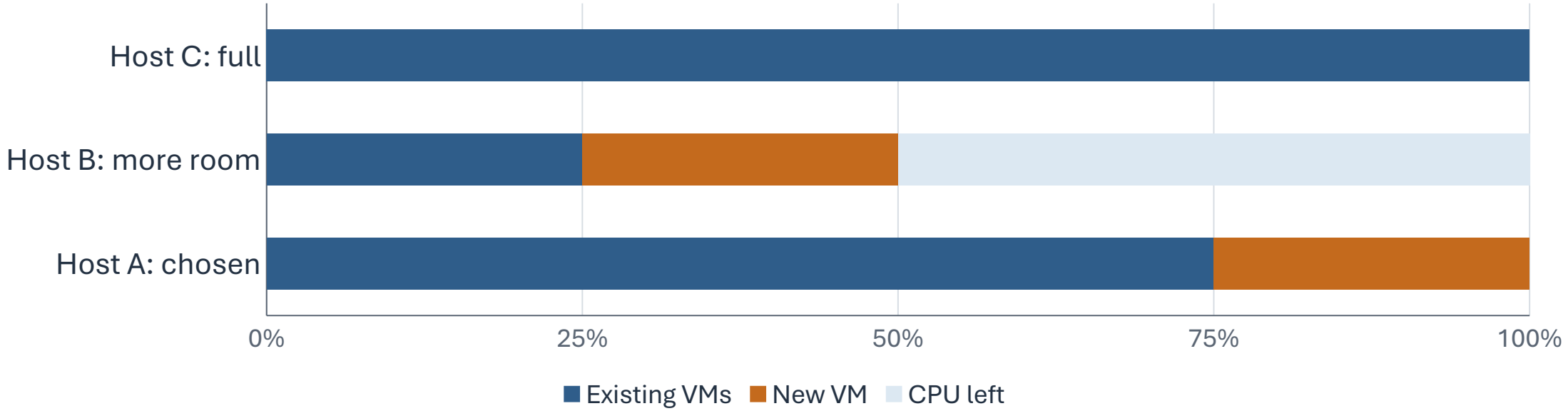


Illustrative capacity values

Best-Fit: pack into the tightest feasible space

Baseline means the method we compare everything against.

New VM: 25% CPU and 12.5% memory

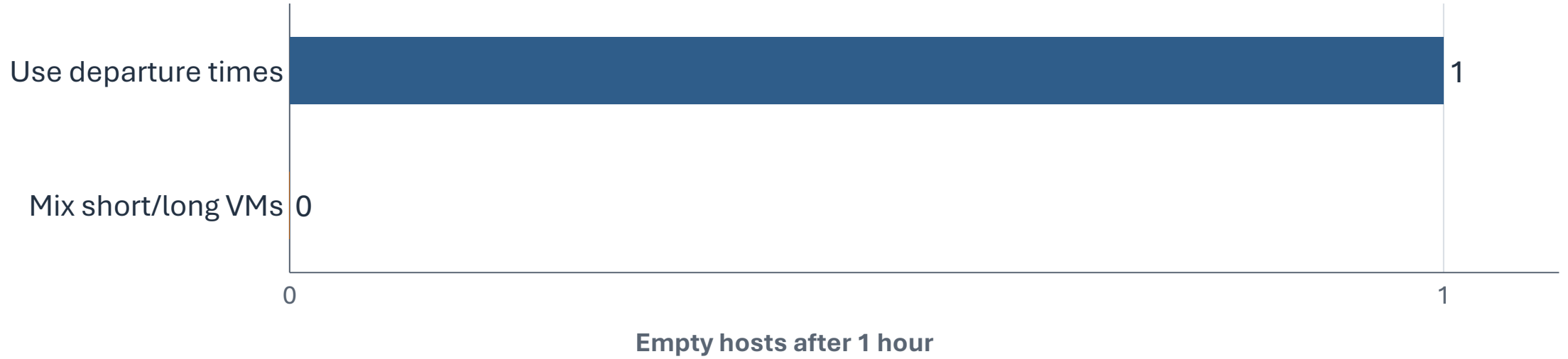


A is the tightest feasible fit. C has no CPU space for the new VM.

Why the time until departure matters

LAVA means Lifetime-Aware VM Allocation.

Same four VMs: two finish after 1 h, two after 10 h. Each host fits two.

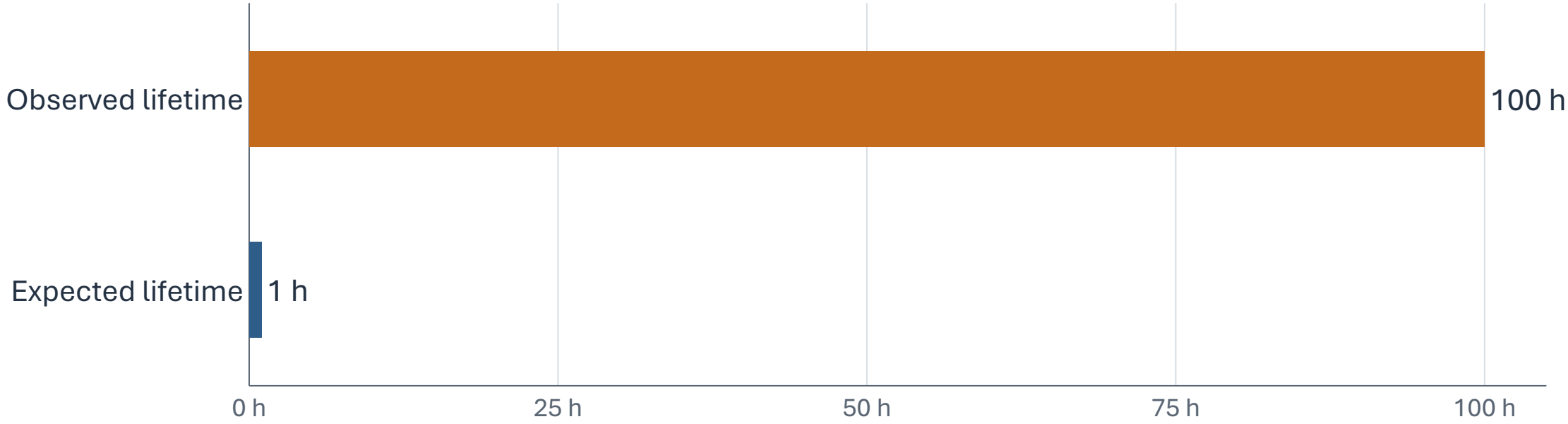


One host can empty earlier when departures line up.

Illustrative data, not the full LAVA rule. LAVA also uses host states and re-prediction.

A wrong short prediction can delay an empty host

A long-running VM can keep the host busy after the others leave.



The host remains occupied much longer than expected.

Illustrative example, not an experimental trace

A fair comparison uses the same incoming VMs

We simulate each workload once, then give it to every scheduler.

Same requests and predictions

- **Best-Fit**
- **LAVA**
- **LAVA with the new predictor**

180
physical hosts

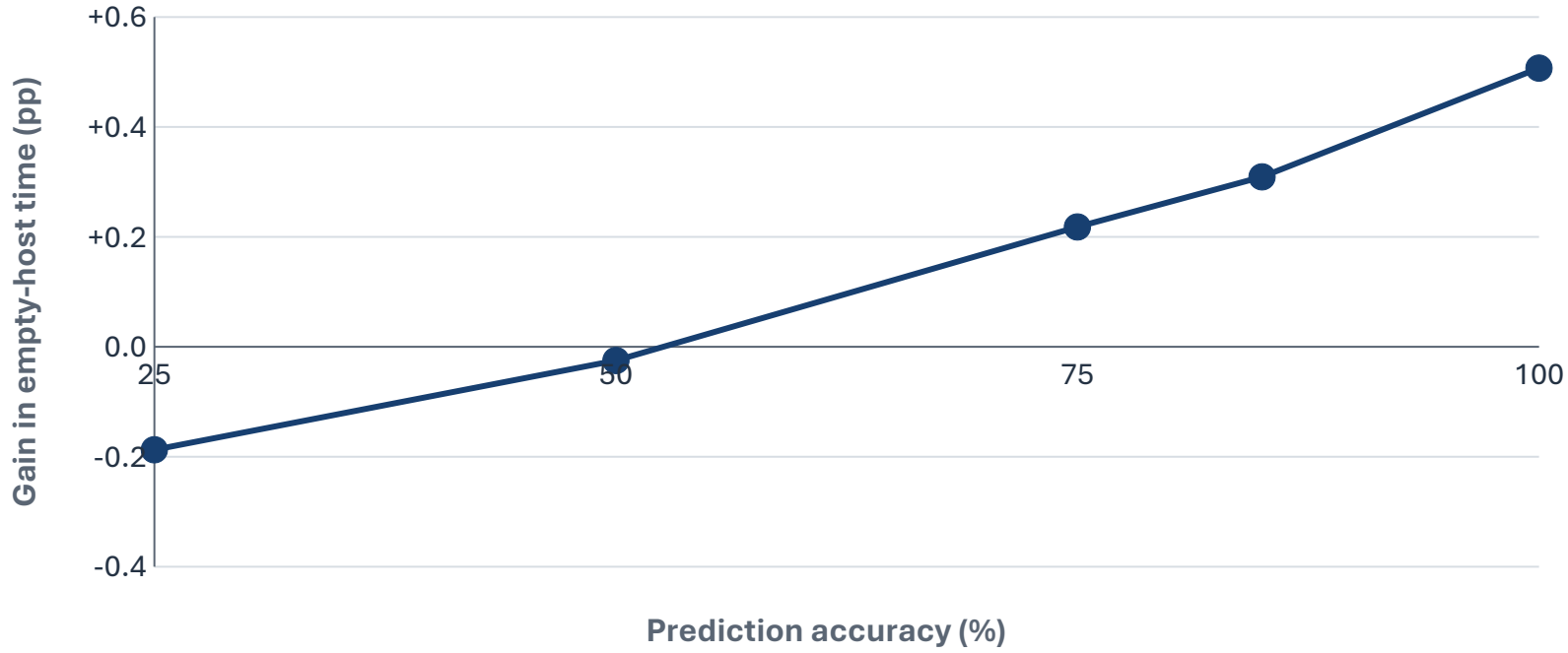
2,500
VMs per workload

100
new test workloads

Goal: more time with completely empty hosts, while admitting every VM.

Poor lifetime predictions can make LAVA worse

Gain over Best-Fit on 100 new test workloads



25% accuracy

-0.187 pp

100% accuracy

+0.507 pp

Accuracy = probability of a near-correct estimate in this controlled test.
pp: 10.0% to 10.2% empty-host time is +0.2 percentage points.

Can confidence choose the scheduler?

Confidence is the probability of the estimated lifetime group.

Use LAVA if confidence is at least 0.3. Otherwise use Best-Fit.

VM-level routing

Each arrival chooses a rule

LAVA or Best-Fit

One shared set of hosts

Host-level routing

Each arrival chooses a pool

LAVA pool or Best-Fit pool

Keep the two modes separate

Routing does not solve the problem here

1 **The trust score can be misleading**

A usually short-lived profile gets low confidence.

2 **The gate can miss the bad estimates**

Under controlled noise, almost every request passes.

3 **Separate pools reduce packing freedom**

An empty slot in the other pool may go unused.

Host-CA minus Best-Fit: -0.090 pp (default predictor)

Use usual behavior to judge a noisy prediction

Bayesian fusion combines prior knowledge with a new estimate.

Usual behavior	The noisy estimate	Updated estimate
This profile is usually short-lived	300 h	1.52 h

Typical profile mean: 1.07 h

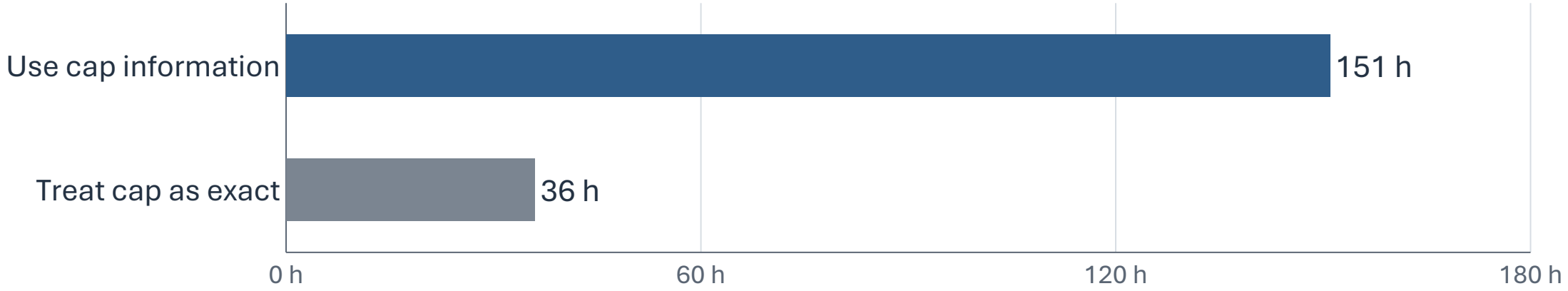
Only the predictor changes. The scheduling rules stay the same.

A VM that is still running gives new evidence

We update the estimate as uptime increases.

The 336 h limit hides how large the noisy estimate was.

Example at 100% accuracy: capped at 336 h; already running for 300 h

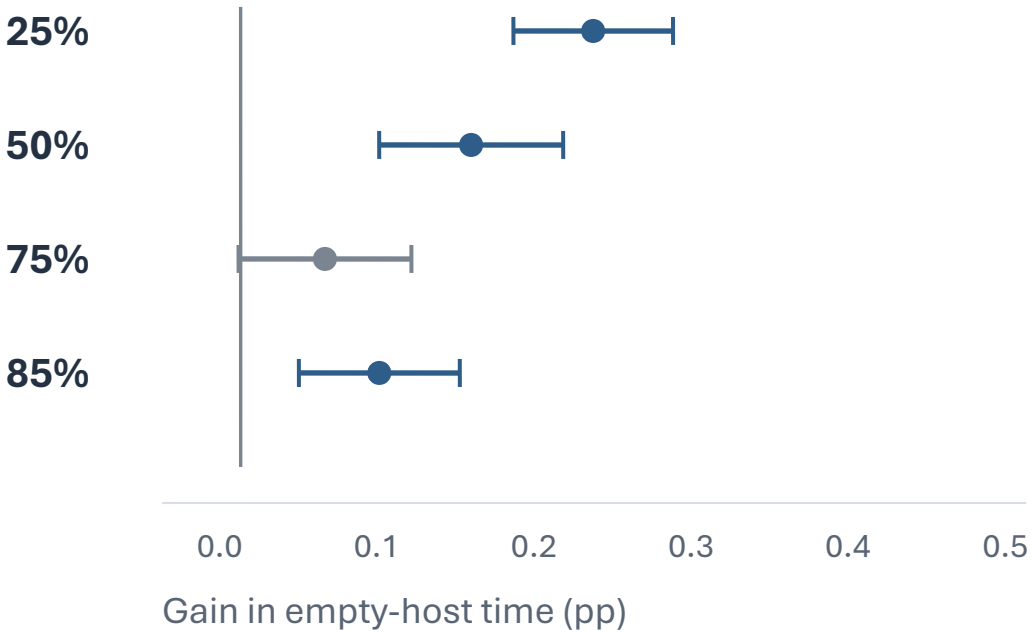


The updated estimate can allow a longer remaining life.

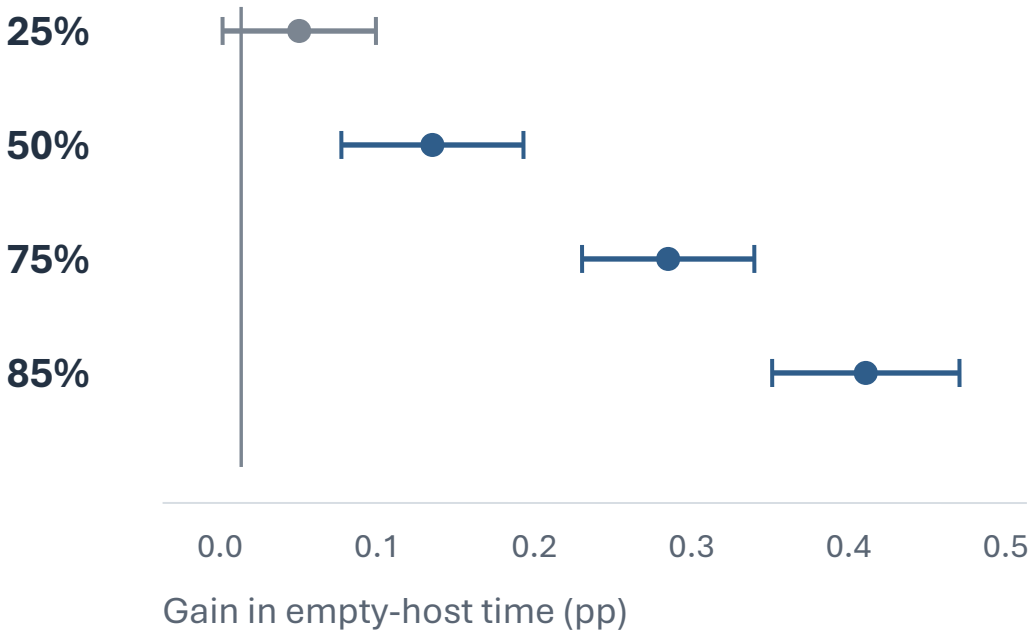
Prior fusion wins 3 of 4 tests against LAVA

The registered variant uses the fused predictor and confidence gate.

Compared with LAVA



Compared with Best-Fit



Gray = not confirmed: 75% vs LAVA, and 25% vs Best-Fit.

The estimate appears to carry most of the gain

Later analysis separates better prediction from the routing gate.

Same fused predictor

Compare it with and without the confidence gate.

Extra effect of adding the gate

-0.018 to +0.011 pp

No clear extra gain from the gate in these tests.

Exploratory analysis, not a registered claim

A prior that is too short can erase the benefit

A wrong prior can discount a correct long-lifetime estimate.

Prior means cut in half

85% prediction accuracy

-0.116 pp

compared with LAVA

Real priors need historical data and checks for changing workloads.

One exploratory robustness case, 50 test workloads

What should we remember?

Findings

Poor predictions can hurt LAVA.

Routing did not fix the problem here.

Prior fusion beat LAVA in 3 of 4 planned tests.

Limits

Synthetic workloads, not production traces

Prior and error model assumed known

Local registration; exploratory robustness

Questions?

Code, data and report

github.com/xxxmian/
CE7490-P2_Revisiting_LAVA